

Quiz (Carolina Reaper)

- Q1.** A shipping company has a fixed cost of \$250 plus an additional \$5 per package. If the total cost is \$315, how many packages are being shipped?
(A) 10
(B) 12
(C) 13
(D) 14
(E) 15
- Q2.** The time t it takes for a truck to travel from City A to City B can be modeled by the equation $t = \frac{1}{4}d^2 - 2d + 15$, where d is the distance. If $t = 5$, what is the possible distance d ?
(A) 8
(B) 10
(C) 12
(D) 16
(E) 20
- Q3.** Solve the equation $x^2 + 6x + 8 = 0$ by factoring.
(A) $x = -2, x = -4$
(B) $x = -1, x = -8$
(C) $x = 2, x = 4$
(D) $x = -2, x = 4$
(E) $x = 1, x = 8$
- Q4.** A traffic engineer is designing a new highway interchange. The equation $y = \frac{1}{3}x^2 - 2x - 5$ models the height of the interchange. What is the y -intercept?
(A) -5
(B) -2
(C) 0
(D) 2
(E) 5
- Q5.** A delivery company has a fleet of trucks with an average speed of 60 miles per hour. The equation $d = 60t - t^2$ models the distance traveled. If the distance traveled is \$1200 miles, what is the possible time t ?
(A) 10
(B) 12
(C) 15
(D) 20
(E) 24
- Q6.** The equation $x^2 - 5x - 6 = 0$ has solutions $x =$?
(A) $x = -1, x = 6$
(B) $x = 1, x = -6$
(C) $x = -2, x = 3$
(D) $x = 2, x = 3$
(E) $x = -3, x = 2$
- Q7.** A logistics company is routing packages through a network of distribution centers. The time it takes to travel from center A to center B can be modeled by the equation $t = \frac{1}{2}d^2 + d + 2$. If $t = 11$, what is the possible distance d ?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q8.** The equation $x^2 + 2x - 15 = 0$ has solutions $x =$?
(A) $x = -5, x = 3$
(B) $x = 1, x = -15$
(C) $x = -3, x = 5$
(D) $x = -1, x = 15$
(E) $x = 5, x = -3$

Open Ended Questions (FRQ)

- Q9.** A package delivery company has a flat rate of \$10 plus an additional \$2 per mile. If the total cost is \$50, write an equation to model the distance traveled and solve for the distance. Show all your work and explain your reasoning.
- Q10.** The height h of a projectile launched from the ground can be modeled by the equation $h = -\frac{1}{4}t^2 + 20t$, where t is time in seconds. If the projectile lands on the ground after t seconds, write an equation to model the time of flight and solve for t . Show all your work and explain your reasoning.

Chef's Tips

- Make sure to check units of measurement
- Graphing calculators are not allowed
- Show all your work and explain your reasoning